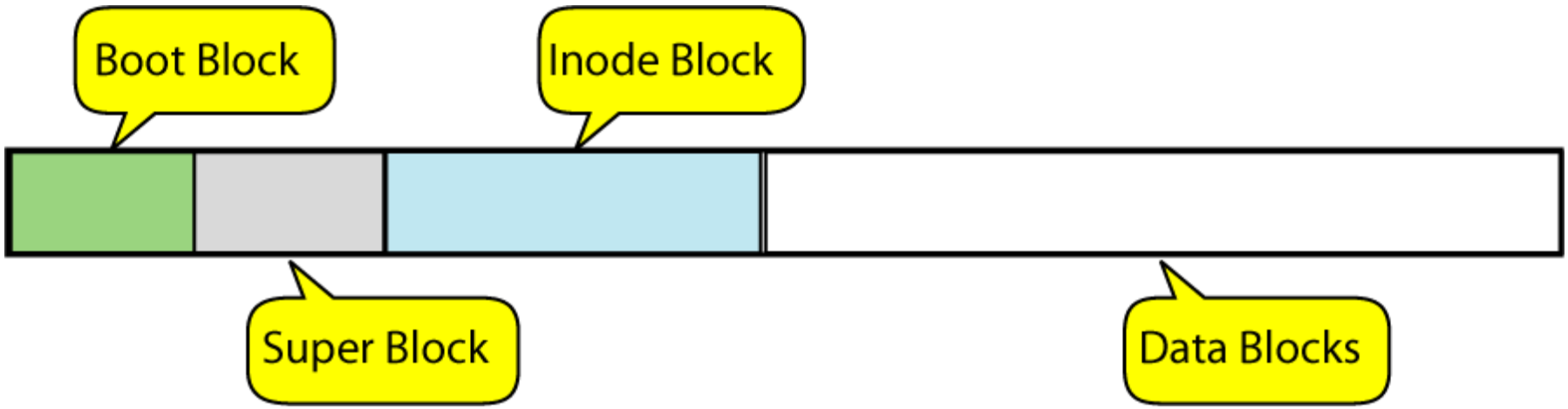
ΜΥΗΠ 228: Προγραμματισμός Συστημάτων
Το Σύστημα Αρχείων (Συνέχεια)

Unix file organization

- Computer has one or more physical hard drives
- Hard drive is divided into partitions

- Partition holds file system
  - File system is set of data blocks
  - Data blocks contain
    - general information
    - actual file data
    - directory information

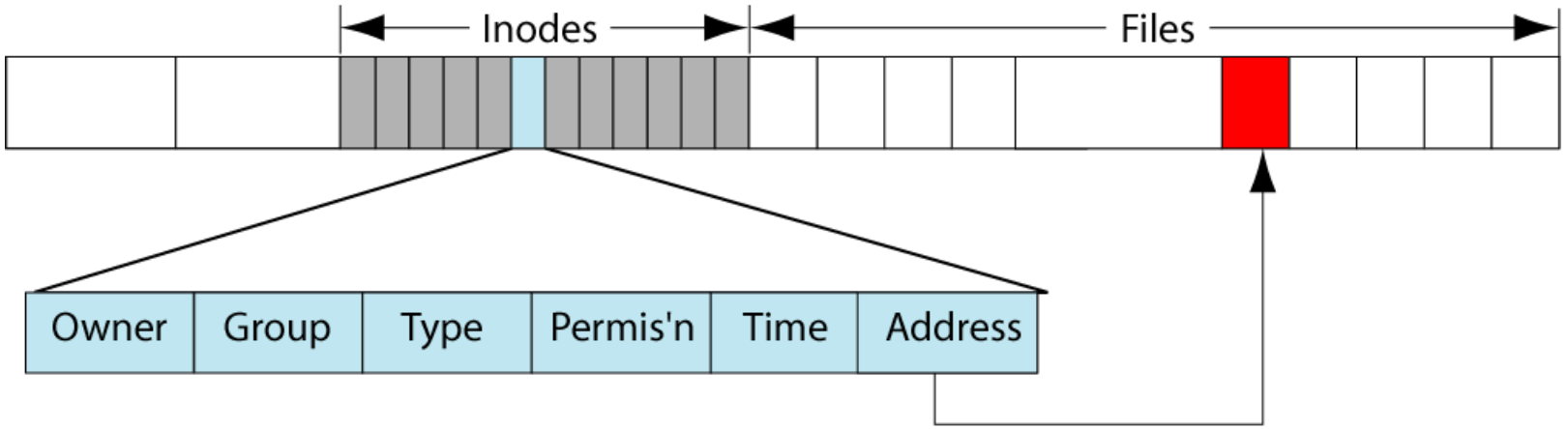
Blocks in a file system



inode

- Index (or information) node: one inode per file
- Each inode has unique number
- contents:
  - File type, access permissions, link count
  - UID, GID
  - Date and time of the file's last
    - **Data access (read and execute)**
    - **Data modification (written)**
    - **I-node modification (permission change)**
  - Data blocks assigned to the file

Inodes in a filesystem



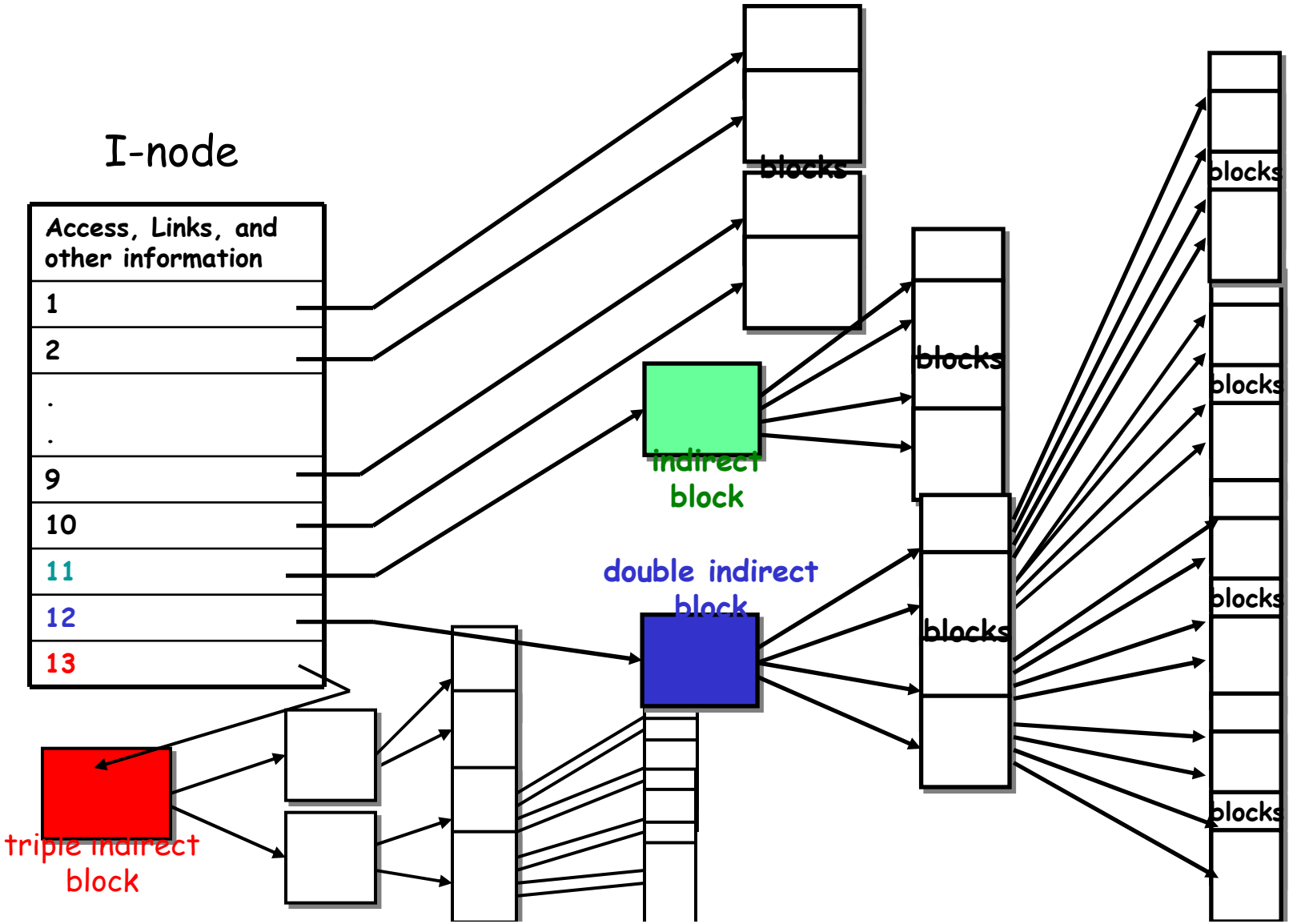
Το μέγεθος των blocks ορίζεται κατά το format του partition

inode Contents: where is the file data ?

Inode may store:

- 10 addresses of data blocks that belong to file
- 1 address of a block that contains data block addresses
- 1 address of a block that contains addresses of blocks that contain data block addresses
- 1 address of a block that contains addresses of blocks that contain addresses of blocks that contain data block addresses

I-node Structure



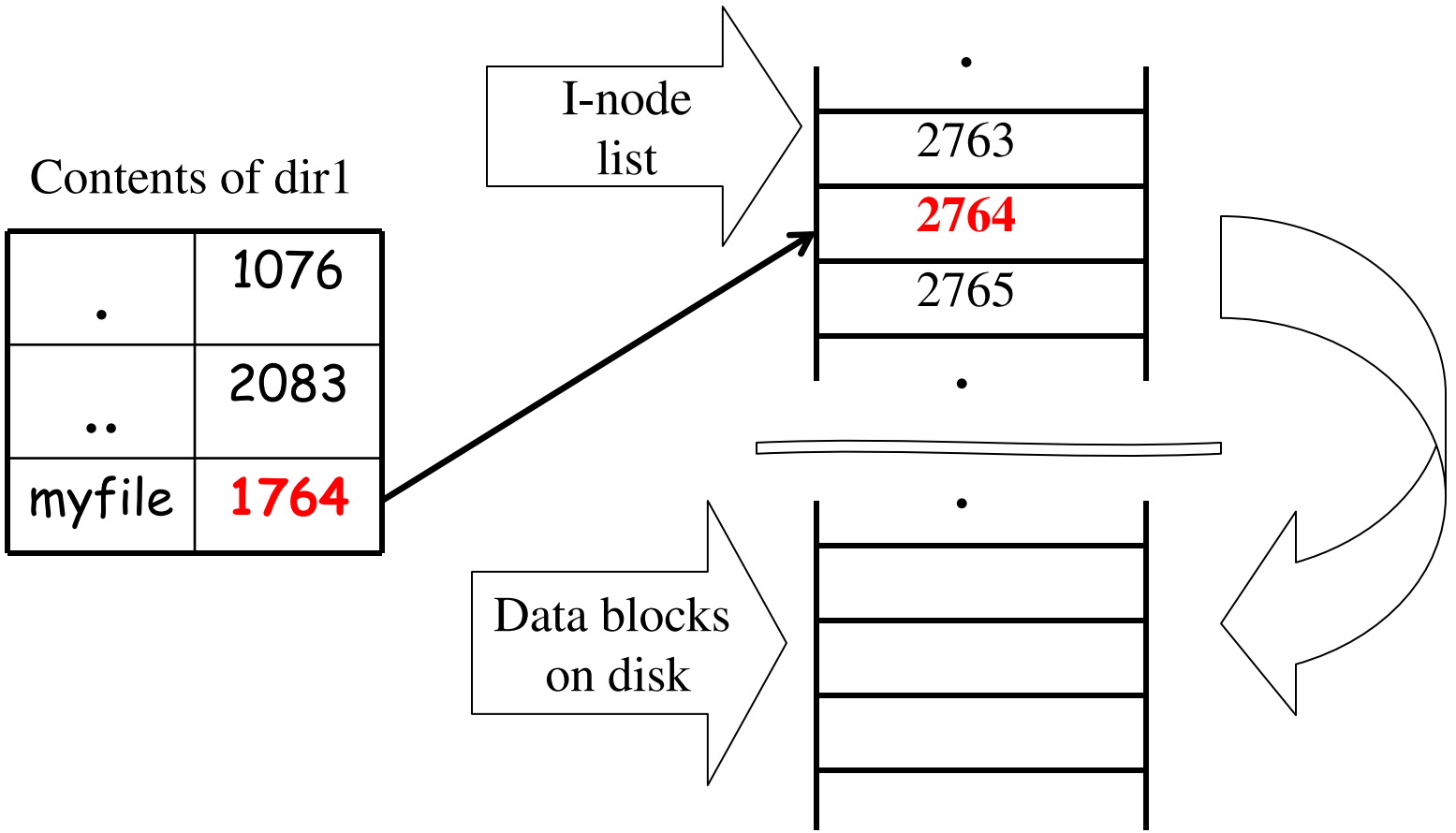
Directory representation

Directory is a file:

- Has inode like regular file, but different file type
- Data blocks of directory contain simple table:

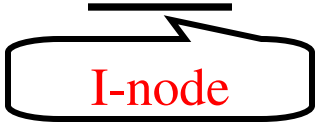
Name	Inode number

Example structure



Output: ls -li

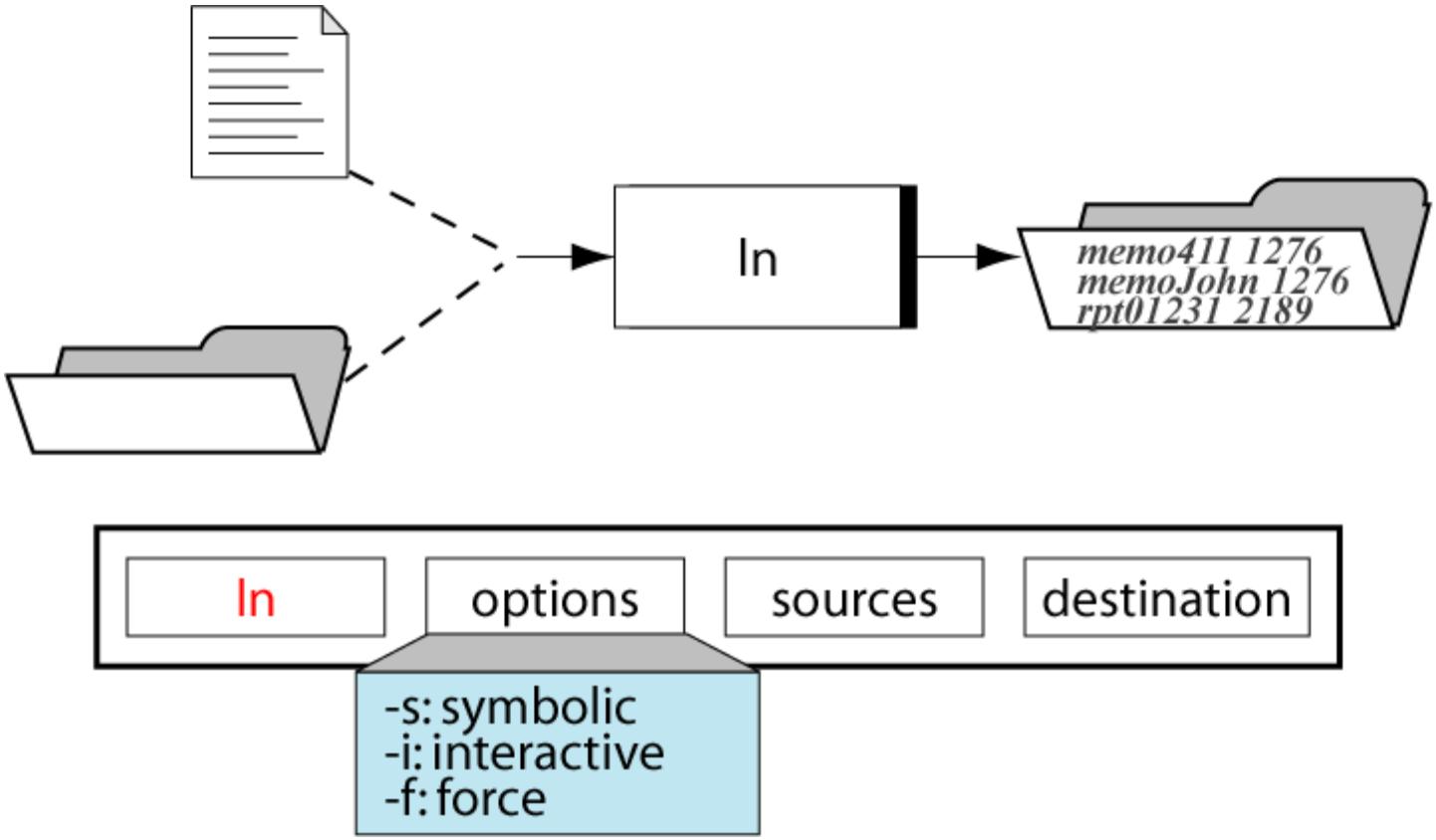
```
ls -lia
total 368
 4249507 drwxr-xr-x   18 bob      it       4096 Oct 10 12:53 .
12583040 drwxr-xr-x   43 root     sys      4096 Oct 20 14:24 ..
 4249509 -rw-r--r--    1 bob      it       3982 Dec 22  2002 .cshrc
 4249510 -rw-r--r--    1 bob      it        631 Dec 13  2002 .login
 4943397 -rw-r--r--    1 bob      it          6 Dec  5  2004 .mailboxlist
 8421158 drwxr-xr-x    2 bob      it         27 Dec 31  2002 BAKS
13400588 drwx------    2 bob      it        130 May 14  2004 DBII
29554097 drwxr-xr-x    2 bob      it        115 May  4  2004 DBII_lab6
 4242824 -rw-r--r--    1 bob      it       3253 Oct 10 12:54 all.txt
   89576 drwx------    2 bob      it         53 Jan 18  2003 courses
 4242823 -rw-r--r--    1 bob      it       2170 Oct 10 12:53 db.txt
 9243983 drwx--x--x    2 bob      it       4096 Oct  7 13:43 dbii_bin
 4510661 -rw-r--r--    1 bob      it      37166 Aug 30  2003 kernel-ntfs-2.4.20-20.9.i686.rpm
```



Linking Files

- To share a single file with multiple users, a link can be used.
- A link is:
  - A reference to a file stored elsewhere on the system.
  - A way to establish a connection to a file to be shared.
- Two types:
  - Hard link
  - Symbolic link (a.k.a. "soft link")

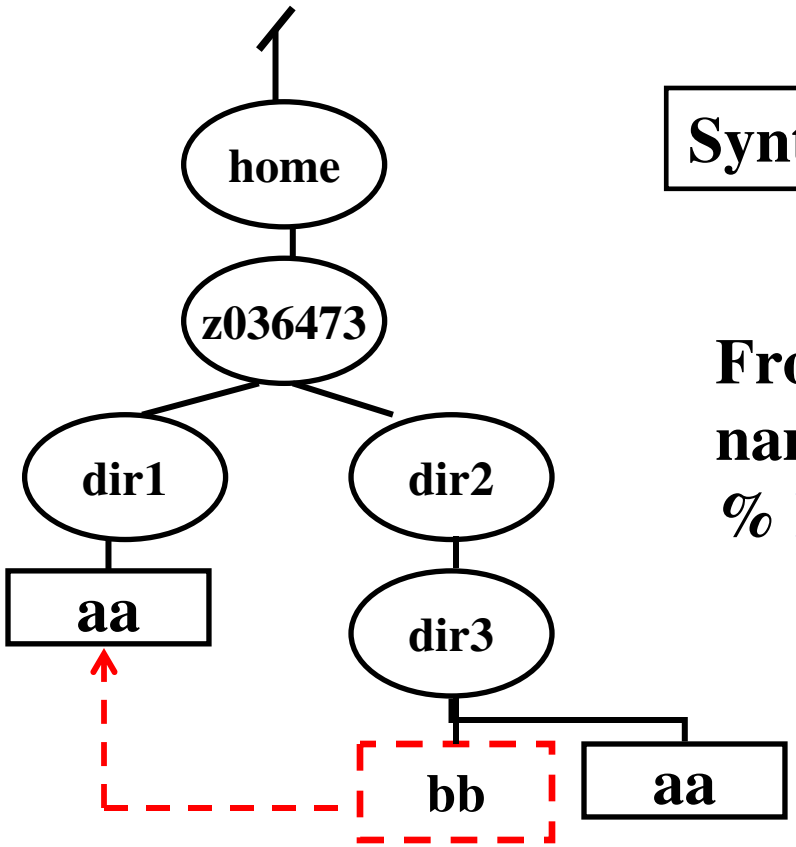
The ln Command



Hard Link

Advantages	Disadvantages
Allow access to original file name via the file name or the I-node number	**Cannot link to a file in a different file system**
The original file continues to exist as long as at least one directory contains its I-node	Prevents owner from truly deleting it, and it counts against his/her disk quota
Checks for the existence of the original file	

Hard Link

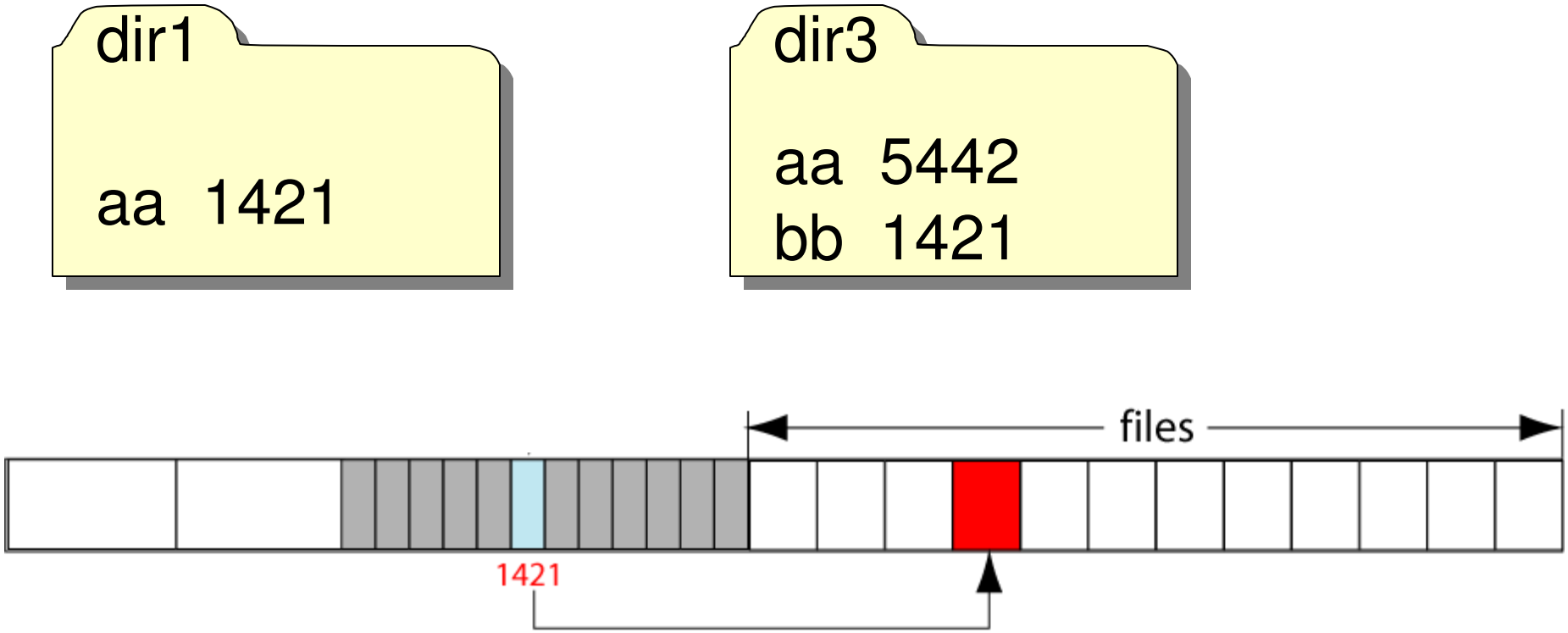


Syntax: ln shared-file link-name

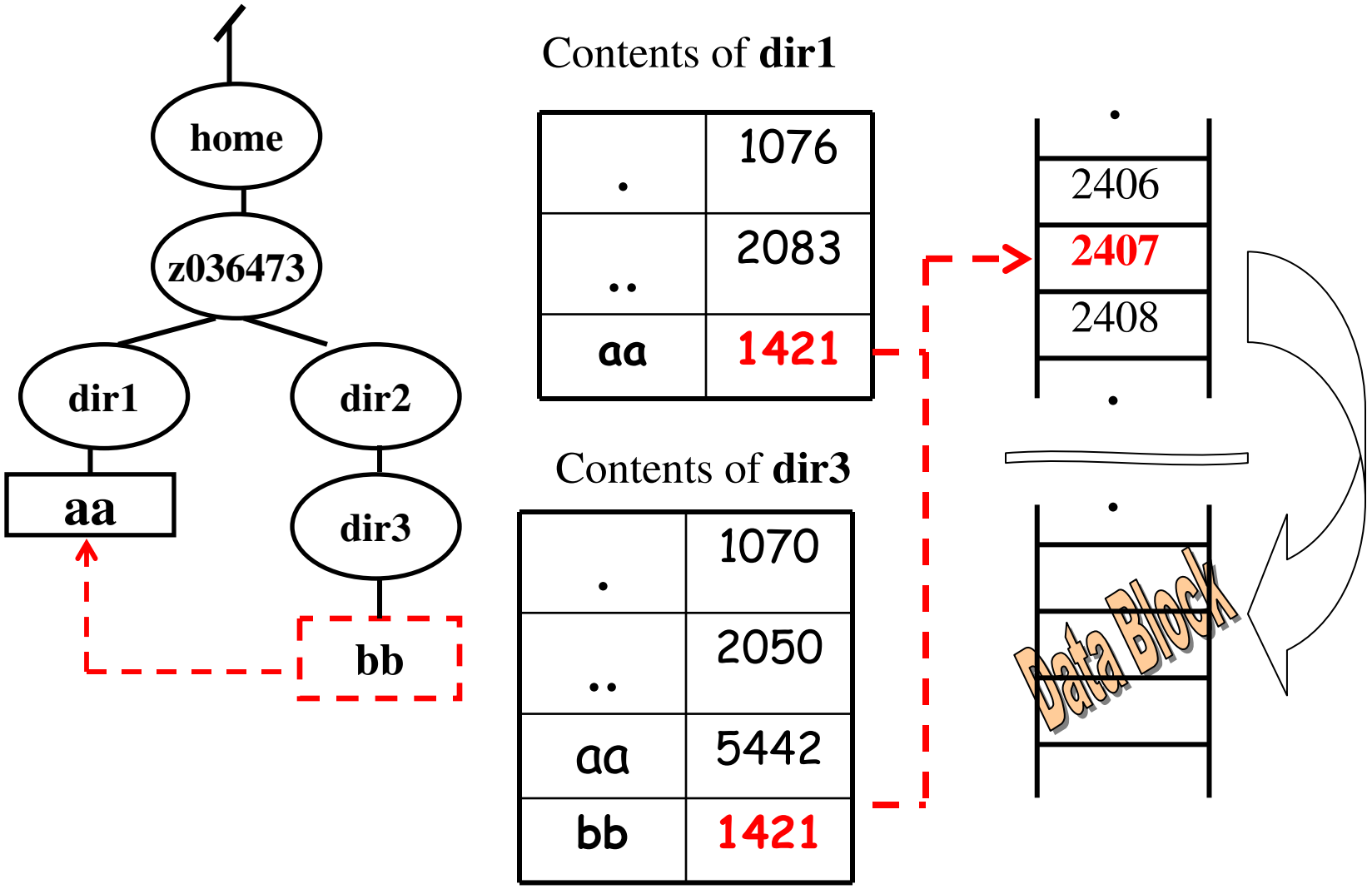
From dir3, link to the file 'aa' in dir1 name it 'bb':

% ln /home/z036473/dir1/aa bb

A Hard Link



Hard Link

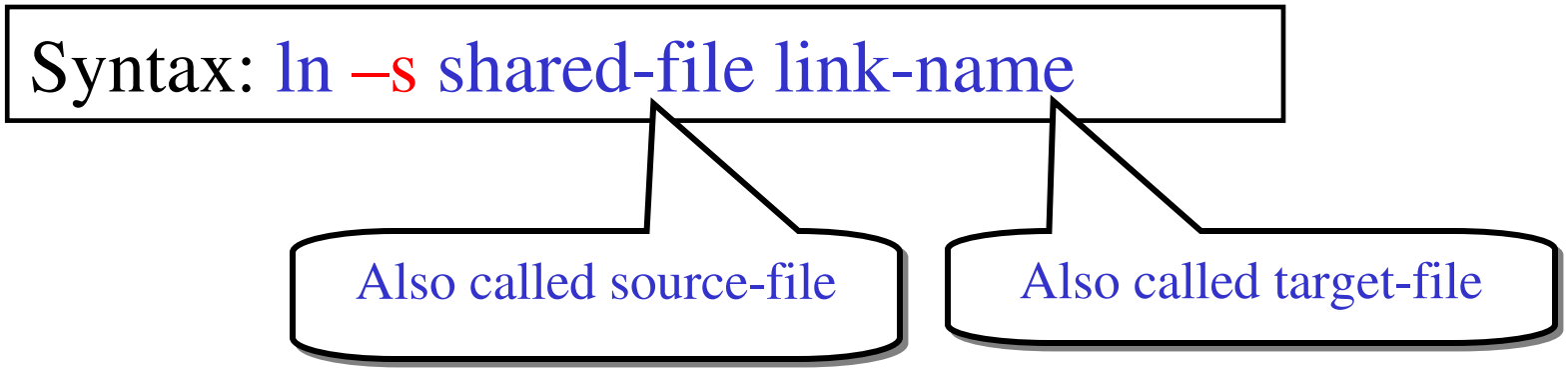

home
z036473
dir1
dir2
aa
dir3
bb
Contents of dir1
.
1076
..
2083
aa
1421
Contents of dir3
.
1070
..
2050
aa
5442
bb
1421
2406
2407
2408
Data Block

Symbolic Link

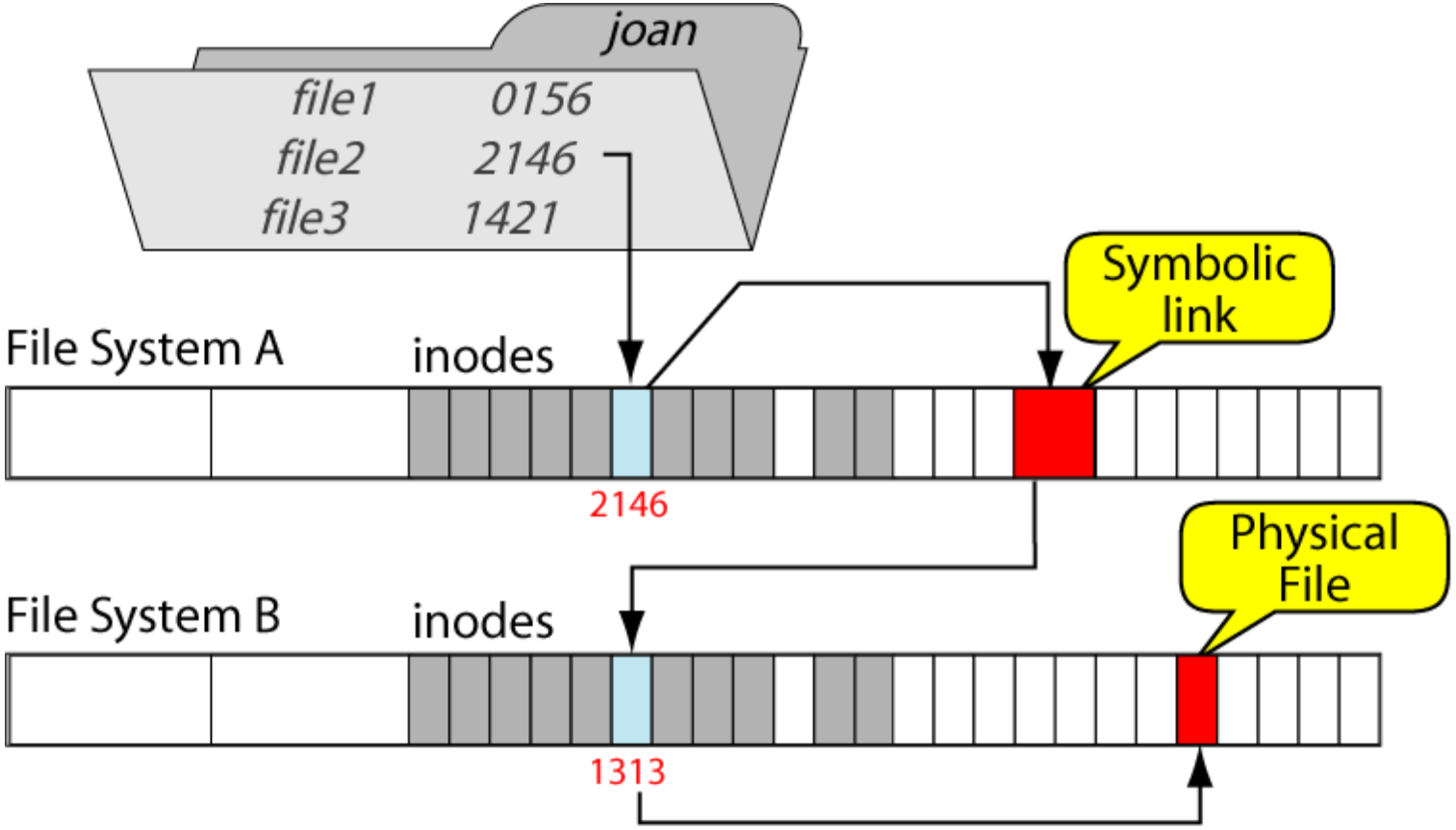
Advantages	Disadvantages
Allow access to original file name	Created without checking the existence of the shared file
Can use either relative or absolute path to access the original file	Cannot access the shared file if its path has restricted permissions
Can cross partition and drives	Can be circular linked to another symbolic linked file
Allows the creation of a link to a directory	

Symbolic Link

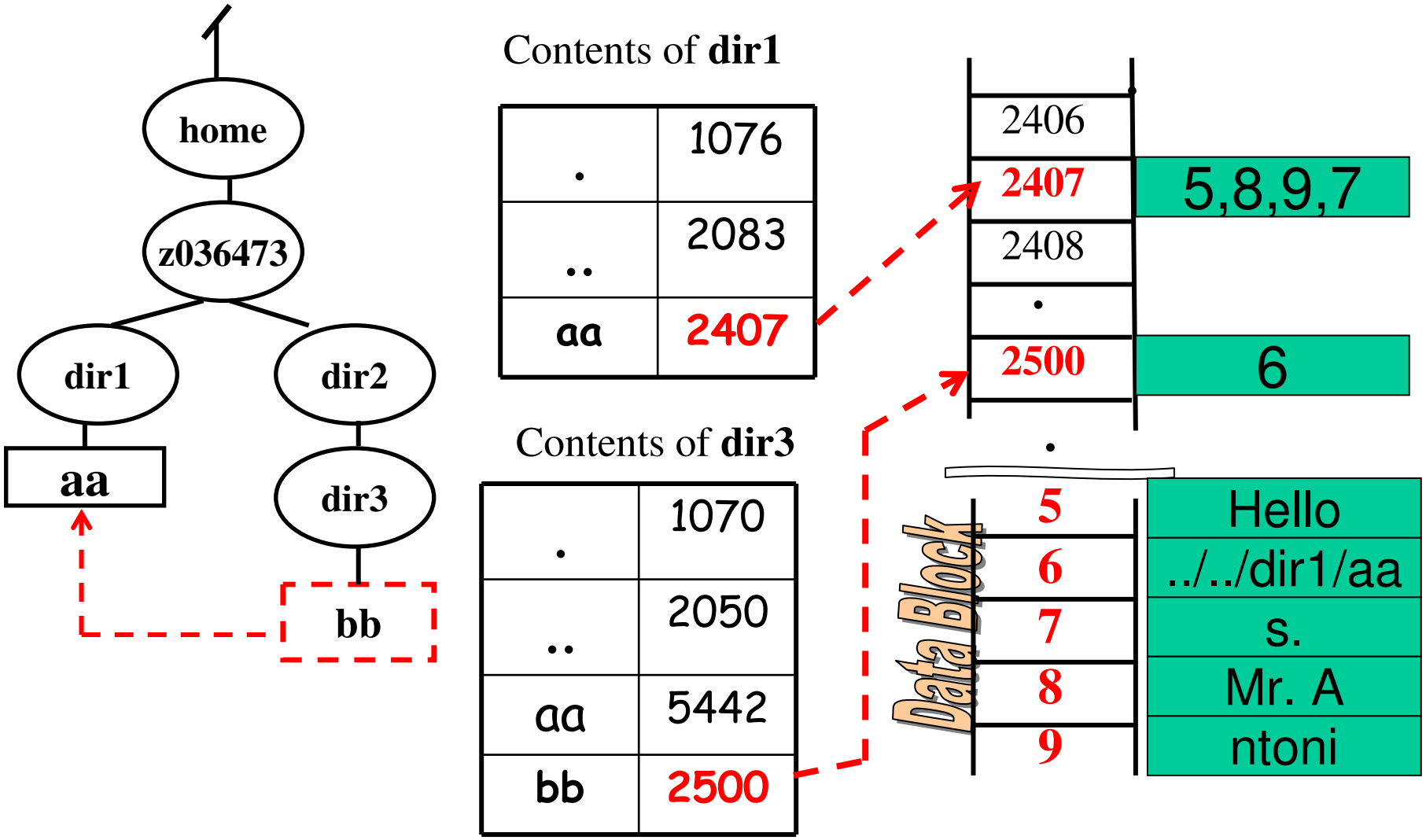
- A hard link may not be created for a file on a different file system
- Use symbolic link
- The linked files do not share the same I-node number



Symbolic Links to Different File Systems



Symbolic Link



Symbolic Link

Εάν προσπαθήσουμε να δούμε τα περιεχόμενα του bb οι αριθμοί δείχνουν με ποια σειρά θα διαβαστούν δεδομένα από τον δίσκο.

